



Achieving enhanced interfacial adhesion and highly oriented structure in PA6/Graphite composites for excellent tribological performance

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ABSTRACT

Friction and wear properties of polyamide (PA) materials can be improved by graphite (Gr) due to its super-lubricity. For further enhancing the ultimate properties of composites, the interfacial adhesion between Gr and PA has already attracted growing attention, while the contribution of Gr arrangement is often ignored. In this work, the interfacial adhesion was improved by π - π interactions between Gr and 1,3-Bis(4,5-dihydro-2-oxazolyl) benzene (BOB) functionalized PA6 via reactive extrusion. Moreover, the highly oriented structure of Gr was achieved by the strong extensional and shearing field. Experimental results indicated that BOB functionalized PA6 with 20 wt% highly oriented Gr (PA6-BOB/HOGr20) displayed excellent balanced performance. Compared with pure PA6, the coefficient of friction (COF) and specific wear rate of the PA6-BOB/HOGr20 composite were reduced by nearly 42% and 46%, respectively. Furthermore, its in-plane thermal conductivity (TC) reached as high as 2.14 W/(m K). Due to the low COF and high TC, the tribo-surface temperature was consistently lower than its glass transition temperature (T_g) under harsh pressure-velocity (PV) conditions of this experiment. In addition, the PA6-BOB/HOGr20 composite still has high ductility, and the elongation at break reaches 150%. This work provides a facile and environmentally friendly method for designing high-performance wear-resistant composites, which can be used as linear guide rail, liner, and linear bearing in industrial applications.

1. Introduction

Greater demands for high-performance wear-resistant polymer materials with exceptional thermal conductivity (TC) and mechanical properties are required to save energy and reduce CO₂ emissions [1–4]. Among the polymer materials, polyamide 6 (PA6) has been used extensively in industry because of the considerable specific strength and excellent chemical resistance [5–8]. But the application of unreinforced PA6 is limited due to its low wear resistance [9,10]. Generally, the tribological properties and heat dissipation ability of PA6 can be improved by adding fillers such as graphite (Gr) [11], graphene (GN) [12], nanodiamond (ND) [13], and carbon nanotubes (CNTs) [14]. However, the poor interfacial adhesion between the fillers and the PA6 matrix limits further improvements in the performance of the composites.

Various surface modification methods of fillers have been proposed

to enhance interfacial adhesion, which serves as a bridge of stress transfer and heat conduction. Zhou et al. modified the basalt fiber (BF) surface by directly coating graphene (GN) via polyurethane sizing agent, which could improve the interfacial adhesion between BF and the PA6 matrix [15]. Compared with BF/PA6, the coefficient of friction (COF) and wear rate of 0.5 GN-BF/PA6 composites decreased by 14.6% and 37.4%, respectively. Li et al. prepared monomer casting nylon 6 (MC PA6) composites reinforced with graphene oxide (GO). The GO surface was treated with aliphatic hydroxyl-terminated poly(dimethylsiloxane) (AHPDMS), which greatly enhanced the interfacial adhesion between GO and MC PA6 [16]. The COF and specific wear rate of the MC PA6/GO composite decreased by 58% and 49%, respectively. Ye et al. synthesized 3-aminopropyl-terminated poly(dimethylsiloxane) (APDMS)-GN/MC PA6 composite in situ through the anchoring effect of APDMS onto the GN surface, revealing a strong enhancement of compatibility in the system [5]. The COF and the specific wear rate were reduced by over

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81% and 48%, respectively. In addition, the interfacial adhesion can also be promoted by introducing functional groups with a high chemical affinity toward fillers into polymer chains [17,18]. As one classic chain extender, bisoxazolines can react quickly with carboxyl groups on PA6 via reactive extrusion [19]. Due to the presence of benzene rings, bisoxazolines modified PA6 should have an affinity with carbon-based fillers through π - π interactions. Therefore, the interfacial adhesion can be enhanced by bisoxazolines modified PA6, which holds promise for further improving the tribological properties of PA6/carbon-based fillers.

For high-performance wear-resistant polymer materials, the heat dissipation capacity is another essential factor, especially under harsh pressure-velocity (PV) conditions. The heat generated during friction will seriously increase wear rate and deteriorate mechanical properties if not being dissipated in time, leading to the failure of composites. In addition to enhancing interfacial interaction and filler dispersion [20, 21], the highly ordered structure of fillers can also significantly improve the heat dissipation ability of composites. Many researchers have designed ordered structures of fillers through different methods, which substantially enhance thermal conductivity. Yang et al. achieved layer structured GNPs/BN in silicon rubber (SiR) via the two-roll mill method, which rendered the composite an outstanding TC of 8.45 W/(m K) at a filler loading of 17.88 vol% [22]. Feng et al. constructed 3D oriented cellulose nanofiber (CNF)/SiC microwires (SiCMWs) networks into the epoxy matrix inspired by the unidirectional freezing casting method [23]. The epoxy composite had a TC of 0.62 W/(m K) with only 1.32 vol % SiCMWs. Gu et al. designed interconnection-oriented graphite nanoplatelets (GNPs) networks in polystyrene matrix based on electrospinning technology [24], resulting in a high TC of 4.72 W/(m K) when the composite contained 35 wt% GNPs. However, most of the reported methods require low viscosity systems and cannot work continuously. In our previous work, a simple and consecutive multistage stretching extrusion (MSE) technique was proposed to introduce highly oriented BN into high viscosity polyethylene melt (HDPE) through the strong extensional and shearing field [25], which had been proved to effectively improve the TC of HDPE/BN composites. Thus, the MSE technique is expected to be used to solve the heat dissipation problem of PA6 composites during friction.

Based on the above literature research, in this work, the synergistic effect of matrix modification and the MSE technique on the tribological properties of PA6/Gr composites will be studied. Firstly, the interfacial adhesion can be enhanced by π - π interactions between Gr and 1,3-Bis (4,5-dihydro-2-oxazolyl)benzene (BOB) functionalized PA6 via reactive extrusion. The oxazoline groups of BOB can quickly react with the carboxyl-terminal group of PA6 to form ester amides through a ring-opening reaction without any byproducts. Secondly, the high orientation of Gr can be achieved by the MSE technique. The friction behaviors, heat dissipation ability, and mechanical properties of PA6/Gr composites will be investigated. This work aims to provide new ideas for designing high-performance wear-resistant PA-based composites.

2. Experimental section

2.1. Sample preparation

PA6 pellets (F136-DH) with a density of 1.13 g/cm³ were purchased from DSM Engineering Plastics company. Graphite flakes (Gr, 10–15 μ m) with a density of 2.2 g/cm³ were obtained from Qingdao Xingyuan graphite milk Co. Ltd. 1,3-Bis(4,5-dihydro-2-oxazolyl)benzene (BOB) with a density of 1.45 g/cm³ was provided by Aladdin Co. Ltd.

PA6, Gr, and BOB were thoroughly dried in the oven and then blended using a co-rotating twin-screw extruder at 250 °C. Afterward, extruded through the custom-made multistage stretching extrusion (MSE) device equipped with several force assembly elements (FAEs) at 250 °C. The MSE technique has been reported in our previous work [26, 27]. Briefly, the molten composites will be divided, extended,

converged, and combined one time through one FAE. The strong extensional and shearing field will improve the dispersion and orientation of Gr, and the processing diagram is shown in Fig. 1. Additionally, samples with or without BOB were prepared by normal extrusion (NE). The name codes and corresponding component proportions are given in Table 1.

2.2. Characterization

SEM (Nova NanoSEM 450) was used to observe the dispersion and orientation of Gr in the PA6 matrix. The samples were cryogenically fractured in liquid nitrogen, and then fractured surfaces were observed. The fracture morphology after stretching and worn surface morphology after friction test were also observed by SEM. X-ray diffraction (XRD) (DX2500 X-ray diffractometer) analysis, using Cu-K α radiation (40 kV, 40 mA, λ = 0.154 nm), was carried out to evaluate the orientation degree of graphite accurately. Lotgering factor f , representing the (00l) orientation degree, was calculated by equation [28]:

$$f_{(00l)} = \frac{P_{(00l)} - P_0}{I - P_0} \quad (1)$$

where

$$P_{(00l)} = \frac{\sum I_{(00l)}}{\sum I_{(hkl)}} \quad (2)$$

$$P_0 = \frac{\sum I_{0(00l)}}{\sum I_{0(hkl)}} \quad (3)$$

where $\sum I_{(00l)}$ and $\sum I_{(hkl)}$ are the sum of (00l) and (hkl) peak intensities, respectively, which appeared in the prepared composites. Similarly, $\sum I_{0(00l)}$ and $\sum I_{0(hkl)}$ are the sum of (00l) and (hkl) peak intensities in the random Gr pattern, respectively. The value of $f_{(00l)}$ is 1 if Gr flakes are completely oriented along in-plane direction.

Thermomechanical properties were performed by dynamic mechanical analysis (DMA, TA Q800) from –10 to 150 °C at a 3 °C/min rate in tension mode with a frequency of 1 Hz and an amplitude of 15 μ m along the extrusion direction. Differential scanning calorimetry (DSC, TA Q200) was used to measure the heat of fusion (ΔH_m) values of samples, the heating temperature ranges from 0 to 250 °C with a ramp of 10 °C/min under a nitrogen atmosphere. The crystallinity (X_c^{DSC}) was calculated by the equation:

$$X_c^{DSC} = \frac{\Delta H_m}{(I - \varnothing)\Delta H_m^0} \times 100\% \quad (4)$$

where ΔH_m is the melting enthalpy of samples, $\varnothing$ is the mass fraction of Gr, ΔH_m^0 is theoretical melting enthalpy of pure PA6, ~230 J/g [29].

The linear reciprocating sliding wear experiment with a ball-on-plate mode (Fig. S1a) was performed on a UMT tribometer (Bruker, Germany). The size of the plate samples is 20 (length) * 20 (width) * 0.8–1 (thickness) mm. Before testing, all the samples were polished with 2000 mesh sandpapers and cleaned ultrasonically in an organic solvent, then dried naturally. The test parameters were shown in Table 2 followed by the ASTM G133-05 standard. The ambient temperature was controlled at 22 $\pm$ 3 °C and the relative humidity was controlled at 50 $\pm$ 5%. Each experiment was repeated three times and the average values were presented with standard deviation. The 3D optical microscope (Contour GT-K, Bruker) was used to measure the cross-sectional area of wear tracks and graphical modeling of wear tracks, as illustrated in Fig. S1b. The specific wear rate was calculated using equation (5) [30].

$$w = \frac{V}{F \bullet L_f} = \frac{A_w \bullet c}{F \bullet L_f} = \frac{\sum_{i=1}^3 A_i \bullet c}{F \bullet L_f} \quad (5)$$

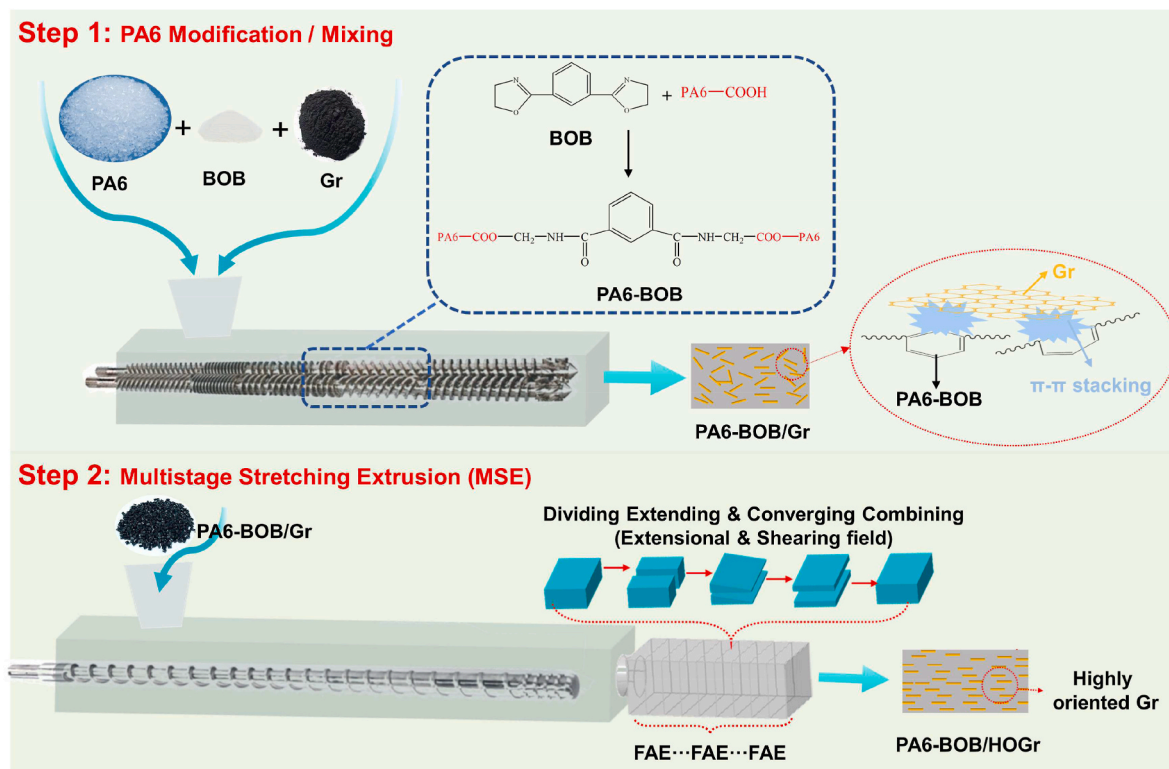


Fig. 1. Schematic illustration of preparing the PA6-BOB/HOGr composites by reactive extrusion and multistage stretching extrusion processing (reaction mechanism for BOB and PA6 is shown inside the blue line). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 1
Component proportions of the prepared Composites.

Name code	PA6 (wt%)	Gr (wt%)	BOB (wt%)
PA6/Gr10	90	10	0
PA6/Gr20	80	20	0
PA6/Gr30	70	30	0
PA6-BOB/Gr10	90	10	0
PA6-BOB/Gr20	80	20	0
PA6-BOB/Gr30	70	30	0
PA6-BOB/HOGr10	89.2	9.9	0.9
PA6-BOB/HOGr20	79.4	19.8	0.8
PA6-BOB/HOGr30	69.5	29.8	0.7

Table 2
Friction test parameters.

Test conditions	value
Material of counterpart ball	304 stainless steel (304-SS)
Diameter of ball	10 mm
Surface roughness of ball: R_a	0.2 μ m
Stroke	10 mm
Sliding distance	100 m
Normal force	10 N
Reciprocating frequency	20 Hz

where w is the specific wear rate ($\text{mm}^3/(\text{N m})$), V is the wear volume (mm^3), A_w is average cross-sectional area of wear track (mm^2), L_f is sliding distance (m), F is normal loading force (N), c is stroke (mm).

TCs (K) of samples were measured as previously reported [25] and calculated by:

$$K = \alpha \times C_p \times \rho \quad (6)$$

where α is the thermal diffusivity, C_p is the heat capacity tested by a laser

flash technique (LFA467, NETZSCH) and ρ is the density of samples. In addition, the tribo-surface temperature during the friction tests was recorded in real time by an infrared thermal imager (T620, FLIR).

The mechanical properties of samples were measured by a universal testing machine (CMT-4104, MTS) with a tensile speed of 50 mm/min at room temperature. Each experiment was repeated five times and the average values were presented with standard deviation.

3. Results and discussion

3.1. Structural characterization

SEM is used to observe the dispersion and orientation of Gr in the PA6 matrix visually. As presented in Fig. 2A1 ~ A3, Gr flakes are agglomerated randomly in PA6/Gr10 ~ PA6/Gr30, and there are apparent gaps between Gr flakes and the PA6 matrix. As can be seen from Fig. 2B1 ~ B3, the addition of BOB results in better dispersion of Gr in the PA6 matrix, and the gaps disappear. Indeed, oxazoline groups of BOB can quickly react with the carboxyl-terminal groups of PA6 to form ester amides without any byproducts in the extrusion process. The reaction mechanism for BOB and PA6 is presented in Fig. 1 [31]. Consequently, the interfacial adhesion is improved by π - π interactions between the benzene rings of PA6-BOB and Gr. In addition, the Gr flakes are highly oriented parallel to the extrusion direction and uniformly distributed in the PA6 matrix through multistage stretching extrusion. The red lines present the visualized explanation and verification in Fig. 2C1 ~ C3, which are attributed to the strong extensional and shearing field provided by FAEs.

To evaluate the orientation degree of Gr in the PA6 matrix accurately, the XRD analysis is carried out. The Gr is a typically anisotropic lamellate filler, the (002) and (100)/(101) peaks at 26° and 41 – 46° correspond to parallel and perpendicular oriented Gr, respectively. The XRD patterns of the composites are depicted in Fig. 3a, and the detailed

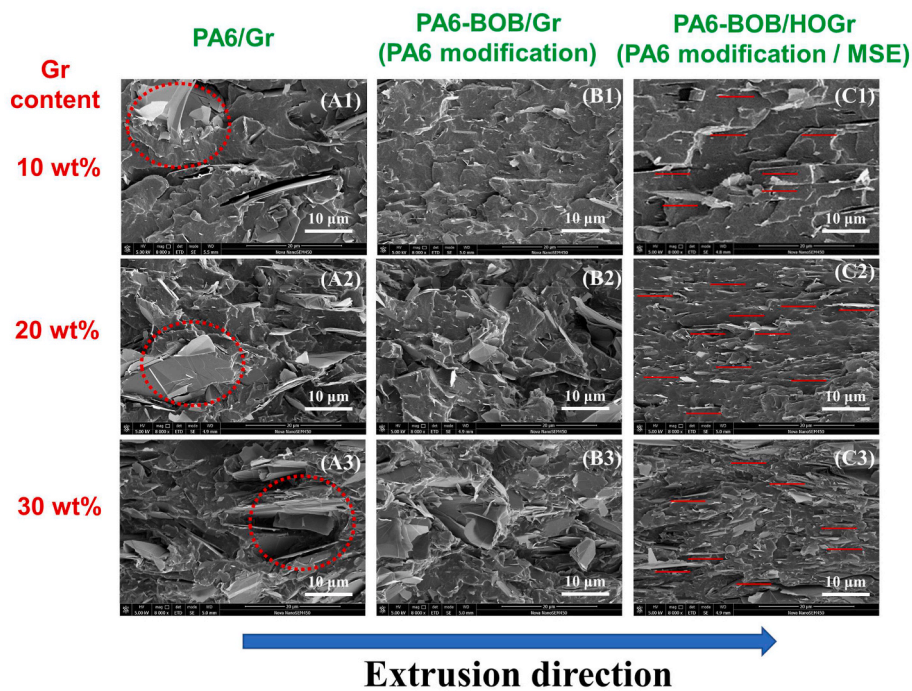


Fig. 2. SEM images of the cryo-fracture cross-section of (A1 ~ A3) PA6/Gr10 ~ PA6/Gr30, (B1 ~ B3) PA6-BOB/Gr10 ~ PA6-BOB/Gr30, and (C1 ~ C3) PA6-BOB/HOGr10 ~ PA6-BOB/HOGr30 (Inside the red circle is aggregated graphite, the red line indicates the direction of graphite orientation). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

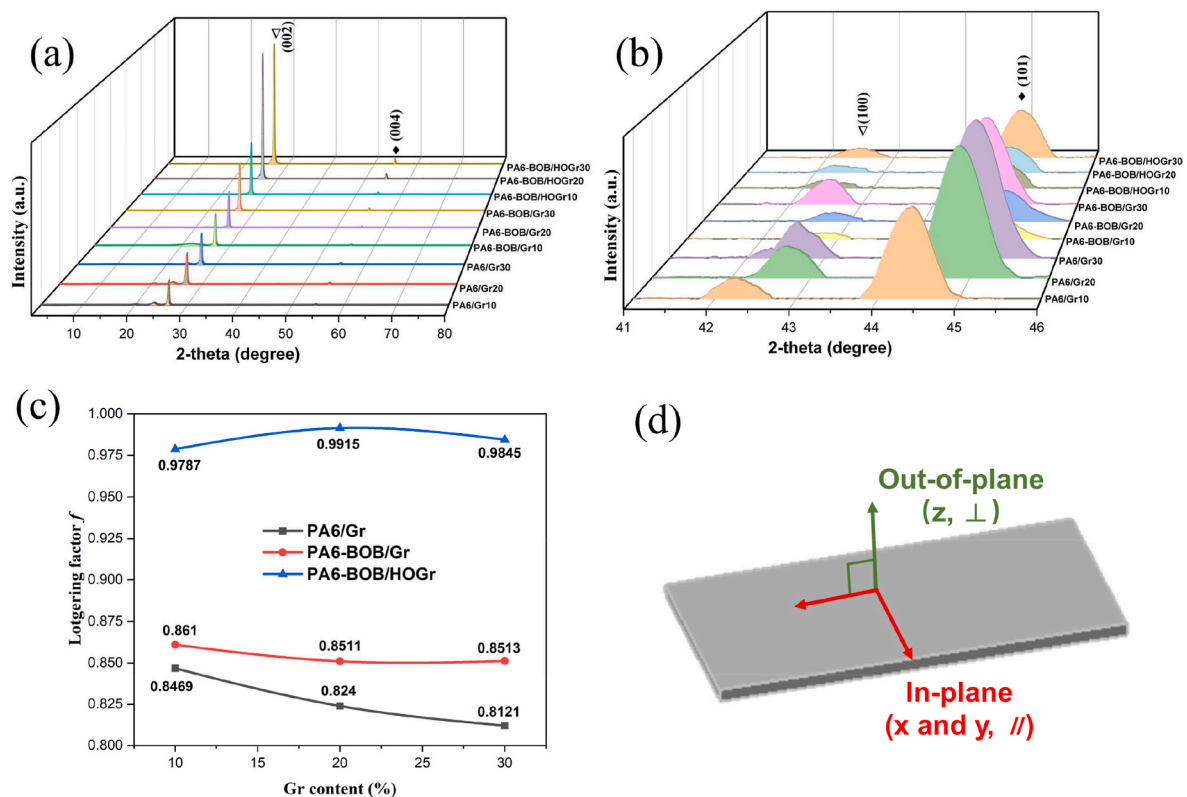


Fig. 3. XRD patterns of the composites in-plane direction (a) from 5–90°, (b) from 41–46° at high magnification, (c) the Lotgering factor f of all composites, and (d) in-plane and out-of-plane directions of extruded composites.

patterns from 41° to 46° are enlarged in Fig. 3b. The detailed patterns appear strong (100)/(101) peaks in the microcrystalline Gr (Fig. S3), which also can be seen in the PA6/Gr composites, whereas weakened in

PA6-BOB/Gr and PA6-BOB/HOGr composites. The Lotgering factor f , representing the (00 l) orientation degree, is shown in Fig. 3c. It is found that the f value of PA6/Gr and PA6-BOB/Gr composites fluctuates

slightly with the increasing loading of Gr and remains at a low level. However, the f value of PA6-BOB/HOGr composites increases sharply. For example, the f value for PA6-BOB/HOGr20, 0.9915, is very close to 1, which means that the (001) crystal plane of Gr is close to perfect orientation parallel to the in-plane direction. The results of SEM and XRD reveal that Gr flakes are uniformly dispersed and highly oriented in the PA6 matrix under the synergistic effect of functionalization of BOB and strong extensional and shearing field of FAEs.

The temperature-dependent dynamic mechanical properties, including storage modulus (E') and damping factor ($\tan \delta$) for pure PA6 and its composites, are presented in Fig. 4 and Table 2. The E' of PA6/Gr composites at room temperature (25 °C) is higher than that of pure PA6 due to the addition of high stiffness Gr, which restricts the mobility of PA6 chain segments, resulting in the “enhancement effect”, and this effect will be enhanced with the increasing loading of Gr. Moreover, the E' of PA6-BOB/Gr composites further increases. The π - π interactions will also restrict the mobility of PA6 molecular chains by tightly binding molecular chains to the surface of Gr. However, the E' of PA6-BOB/HOGr composites decreases slightly with the same loading of Gr, which may be because the restriction on the mobility of PA6 molecular chains is reduced when Gr flakes are highly oriented. The interfacial adhesion can influence the mechanical behaviors of composites significantly, which can be investigated by the interfacial adhesion parameter (β) [32].

$$\frac{(\tan \delta_{\max})_C}{(\tan \delta_{\max})_m} = 1 - \beta \phi_v \quad (7)$$

In Eq. (7), $(\tan \delta_{\max})_C$ and $(\tan \delta_{\max})_m$ represent the values of $\tan \delta$ peak of PA6-based composites and PA6 matrix, respectively; ϕ_v is the volume fraction of Gr and can be calculated from mass fraction and density. Generally, the higher the β value, the stronger the interfacial adhesion. As shown in Table 3, whatever the processing method, the composites containing 10 wt% Gr have the highest β value because the

Table 3

Mechanical data obtained from DMA and crystallinity obtained from DSC.

Sample code	E' (25 °C) (GPa)	T_g (°C) (°C)	ϕ_v (vol) (%)	$\tan \delta_{\max}$	β	X_c^{DSC} (%)
PA6	3.25	60.3	0	0.168	—	29.1
PA6/Gr10	4.50	65.8	5.40	0.1447	2.5683	24.1
PA6/Gr20	5.88	67.1	11.38	0.1440	1.2553	23.6
PA6/Gr30	8.06	68.2	18.04	0.1438	0.7985	23.1
PA6-BOB/ Gr10	4.89	67.7	5.40	0.1411	2.9652	23.7
PA6-BOB/ Gr20	6.34	70.0	11.38	0.1400	1.4646	22.2
PA6-BOB/ Gr30	8.52	70.4	18.04	0.1387	0.9668	22.1
PA6-BOB/ HOGr10	5.94	69.1	5.46	0.1375	3.3250	23.4
PA6-BOB/ HOGr20	6.95	73.8	11.64	0.1331	1.7847	22.1
PA6-BOB/ HOGr30	9.66	74.2	18.70	0.1261	1.3337	22.0

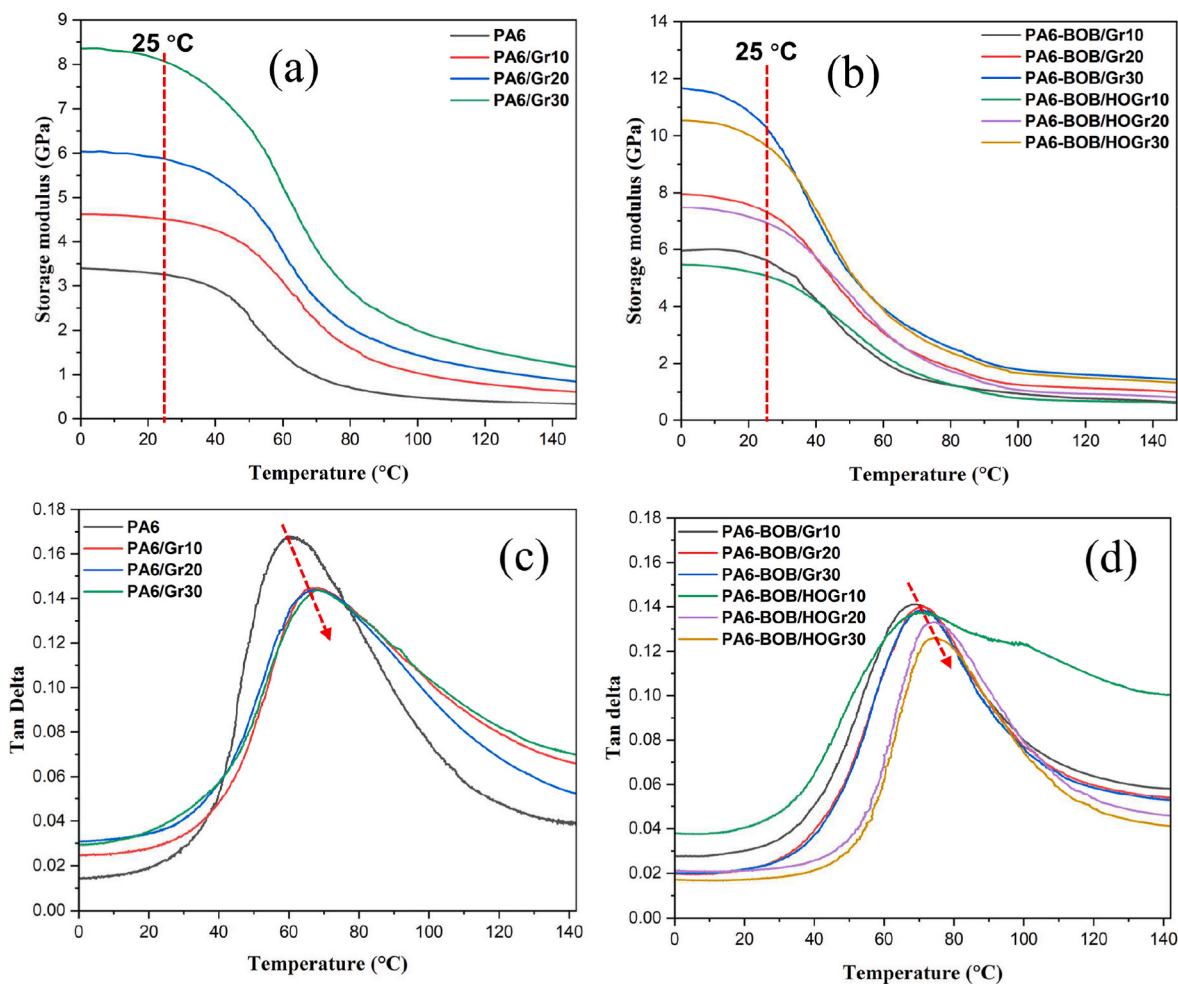


Fig. 4. (a, b) Storage modulus curves and (c, d) tan delta curves of pure PA6 and its composites (the red arrow indicates the offset of tan delta peaks of samples). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Gr does not aggregate seriously at a low loading level, resulting in excellent interfacial adhesion with the PA6 matrix. BOB functionalized PA6 leads to an increment of β value, proving that BOB improves the interfacial adhesion between Gr and PA6 matrix. Moreover, the β values of the composites prepared by the MSE method are higher than those prepared by the NE method at the same loading of Gr, attributed to the more uniform dispersion. In addition, the temperature corresponding to the maximum of the tan delta peak is the glass transition temperature (T_g). It can be pointed out from Fig. 4c and d, and Table 3, the tan delta peaks (T_g) of PA6-BOB/Gr composites shift to a higher value, and those values of PA6-BOB/HOGr composites are highest. This phenomenon is most likely because of enhanced interfacial adhesion due to uniform dispersion and high orientation of Gr in the PA6 matrix.

The crystalline degree of samples calculated through the DSC data is listed in Table 3. The X_c^{DSC} values of PA6 decrease with the increasing loading level of Gr because the mobility of PA6 chains is limited by Gr, and the ability to form crystal structures decreases. The addition of BOB also reduces the crystallinity, attributed to the improvement of interfacial adhesion, further restricting the mobility of PA6 molecular chains and crystal growth.

3.2. Friction and wear properties

The COF curves of pure PA6 and PA6/Gr composites are shown in Fig. 5a, all the Gr reinforced PA6 composites exhibit lower COF than pure PA6. Moreover, the COF of composites reaches a stable value at the initial stage of friction. The Gr shows a good anti-friction performance because the weak van der Waals force between the two-dimensional flakes makes Gr easily interlayer slip during friction [33]. The friction and wear mechanisms are illustrated in Fig. 6A. It should be noted that the PA6/Gr20 composite has the lowest COF and specific wear rate,

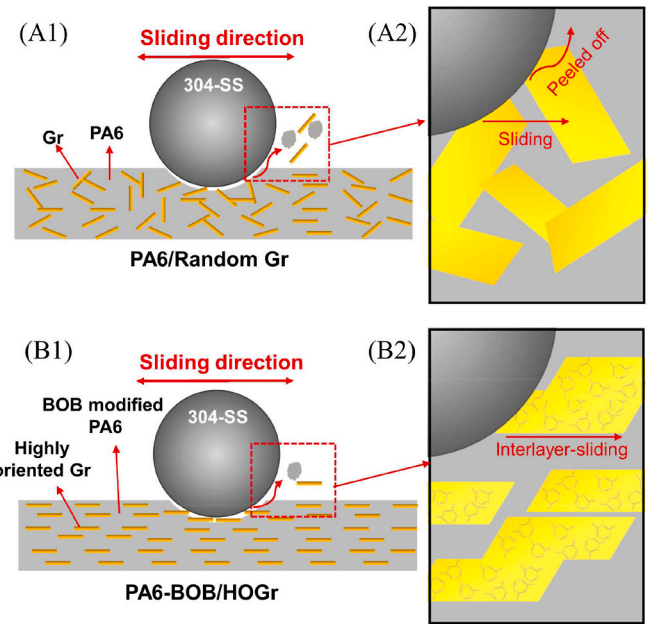


Fig. 6. Schematic diagrams of friction and wear mechanisms of (A1, A2) PA6/Gr and (B1, B2) PA6-BOB/HOGr.

approximately decreased by 32% and 44% compared with pure PA6, respectively. (statistics shown in Fig. 5d and e). Whereas the COF and specific wear rate increase slightly with the increasement of Gr loading, which may be due to the aggregation of Gr at a high loading [11]. According to Fig. 5b, 5d, and 5e, the COF and specific wear rate are

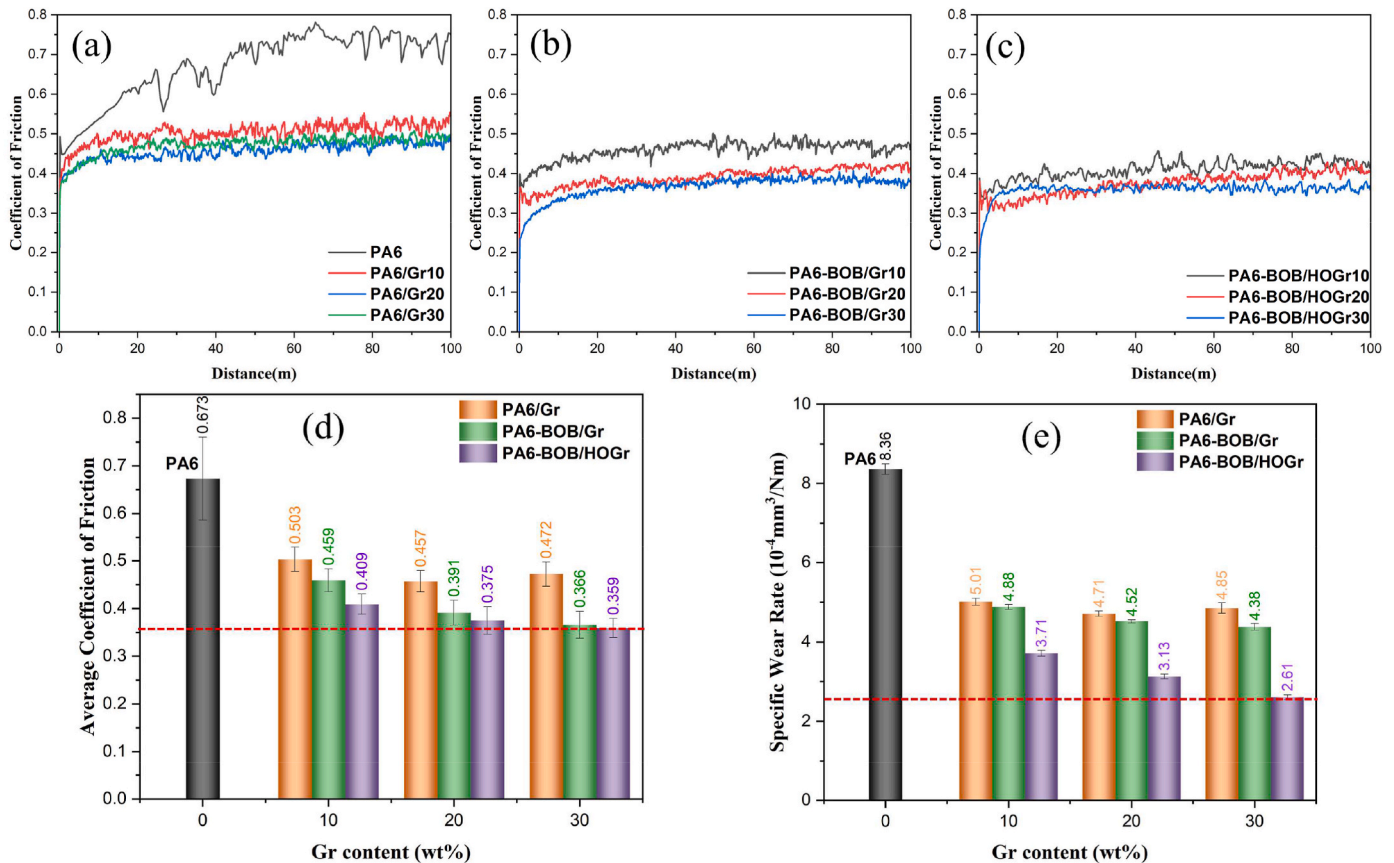


Fig. 5. Coefficient of friction curves of (a) pure PA6 and PA6/Gr10 ~ PA6/Gr30, (b) PA6-BOB/Gr10 ~ PA6-BOB/Gr30, (c) PA6-BOB/HOGr10 ~ PA6-BOB/HOGr30; statistical charts of (d) average coefficient of friction and (e) specific wear rate of pure PA6 and its composites.

considerably reduced with the addition of BOB. The COF and wear rate of PA6-BOB/Gr30 composites are the lowest, 45.6% and 47.6% lower than PA6, respectively. It can be explained as the immobilization effect of π - π interactions, resulting in Gr hardly being peeled off from the PA6 matrix during friction.

The COF and wear rate of PA6-BOB/HOGr composites are further reduced, as shown in Fig. 5c, 5d, and 5e. The composite PA6-BOB/HOGr30 has the lowest COF and wear rate, decreased by 46.7% and 68.8% compared with PA6, respectively. According to the literature [34–36], the friction behavior of Gr flakes is anisotropic, the average COF of the parallel surface of the basal plane (0.1) is lower than that of the perpendicular surface. The friction and wear mechanisms are shown in Fig. 6B, and each highly oriented Gr flake has a large contact area with the PA6 matrix. Meanwhile, benzene rings of PA6-BOB are

anchored on the Gr surface due to π - π interaction. For the above reasons, Gr flakes tend to remain on the surface of the PA6 matrix during friction to form a protective layer and reduce the COF and wear rate.

The worn surfaces of the pure PA6 and some representative composites are investigated by SEM. As presented in Fig. 7A, the surface of the pure PA6 is considerably damaged with apparent ploughing damage and rugged bumps (pointed by red arrows). Many debris appear on the worn surfaces for the composites PA6/Gr20 and PA6/Gr30 (Fig. 7B1 and B2), revealing that Gr flakes are peeled off from the PA6 matrix. For PA6-BOB/Gr20 and PA6-BOB/Gr30, the damage to worn surfaces is barely visible (Fig. 7C1 and C2) due to the strong interfacial adhesion between Gr and PA6, consequently, wear resistance is improved. As expected, the worn surfaces of PA6-BOB/HOGr20 and PA6-BOB/HOGr30 (Fig. 7D1 and D2) display the smoothest characteristics

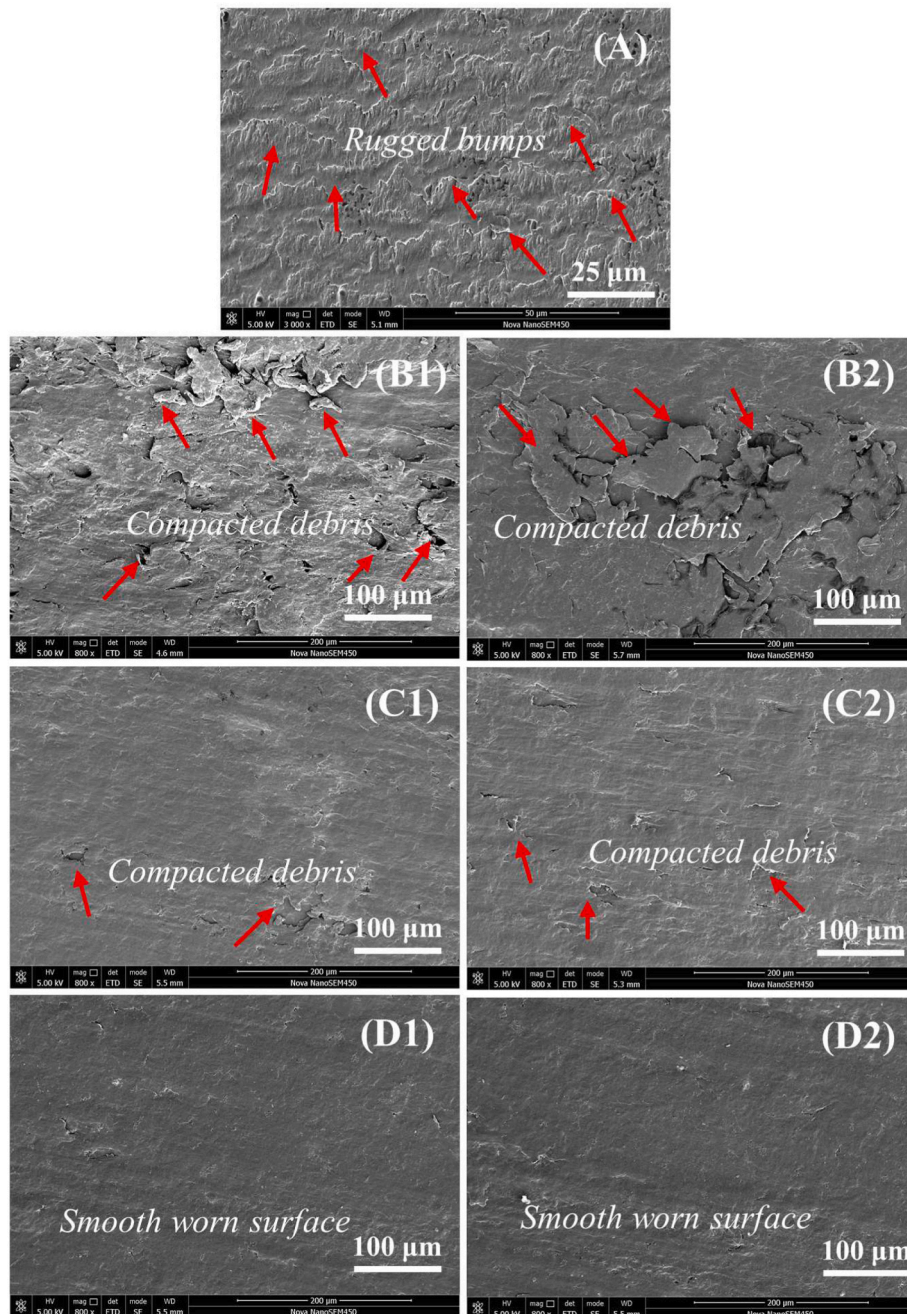


Fig. 7. SEM images of worn surfaces of some representative samples: (A) pure PA6, (B1) PA6/Gr20, (B2) PA6/Gr30, (C1) PA6-BOB/Gr20, (C2) PA6-BOB/Gr30, (D1) PA6-BOB/HOGr20 and (D2) PA6-BOB/HOGr30.

among all the samples. The results indicate that the synergistic effect of enhanced interfacial adhesion and the highly oriented structure of Gr effectively prevented wear damage.

3.3. Heat dissipation performance

As illustrated in Fig. 8a, the in-plane thermal conductivities (TCs) of all samples were measured. The TCs of PA6/Gr composites increase with increasing loading of Gr, the TC of PA6/Gr30 reaches 0.81 W/(m K), which is 2.89 times that of pure PA6. While the value of 1.13 W/(m K) is achieved for PA6-BOB/Gr30, 4.03 times that of pure PA6, which is due to the reduction of phonon scattering by strong interface interaction between Gr and PA6. Moreover, the TC of PA6-BOB/HOGr30 reaches as high as 2.96 W/(m K), which is 10.6 times that of pure PA6. The highly oriented Gr builds the path of phonon transmission and reduces phonon scattering. A high TC value can be achieved even at a relatively low Gr loading. For example, PA6-BOB/HOGr20 has a TC of 2.14 W/(m K). Therefore, making Gr highly oriented is an effective method to improve the TC of composites TC significantly.

To demonstrate that the improvement of TC is beneficial to dissipating the friction heat, the temperature change of the tribo-surface during the friction was recorded by an infrared thermal imager. The maximum temperature of the tribo-surface is shown in Fig. 8b. According to the literature [37], a low COF can reduce the generation of friction heat, while a high TC can quickly dissipate the generated friction heat. For example, the COF of PA6-BOB/HOGr30 (0.359) is similar to that of PA6-BOB/Gr30 (0.366), thus, the generated friction heat should be similar. But the maximum tribo-surface temperature of PA6-BOB/HOGr30 (43.5 °C) is much lower than that of PA6-BOB/Gr30 (55.1 °C), since the TC of the former is 61.8% higher than the latter, which contributes to the rapid dissipation of the friction heat. The

infrared images of some representative samples are shown in Fig. 8c. Clearly, the maximum temperature decreased from 108.3 °C for neat PA6 to 51.4 °C for PA6-BOB/HOGr20, much lower than the glass transition temperature. Consequently, the oxidation degree will be reduced under harsh PV conditions, improving reliability and prolonging the service life of the materials.

To better understand the heat dissipation mechanism, the heat transfer models of normally extruded BOB-free composites (Fig. 9a) and multistage stretching extruded composites with BOB (Fig. 9b) are proposed. Due to the strong phonon scattering of imperfect crystals or other defects, an effective heat conduction pathway cannot be formed, leading to a low TC of polymers [38]. The incorporation of thermally conductive fillers is treated as a feasible method. However, the TC cannot be significantly improved by simply mixing with the polymer matrix because the interfacial adhesion between constituents and filler arrangement in the matrix has a significant influence on the TC. As depicted in Fig. 9a, the randomly dispersed Gr cannot form a better heat conduction pathway in the PA6 matrix. The heat dissipation rate is as slow as a tortoise crawling, mainly due to defects and phonon scattering. However, the synergistic effect of enhanced interfacial adhesion and highly oriented structure of Gr can form a straight heat conduction path, the heat dissipation rate is as fast as a cheetah running.

3.4. Mechanical properties

The typical stress–strain curves of pure PA6 and all composites are shown in Fig. 10a ~ c and statistical data in Fig. 10d ~ e. For PA6/Gr composites, the PA6/Gr20 exhibits the highest tensile strength, increasing approximately 36.5% compared with pure PA6, because the stress transfers to the high stiffness Gr during the tensile test. In contrast, the elongation at break decreases sharply due to the aggregations of Gr

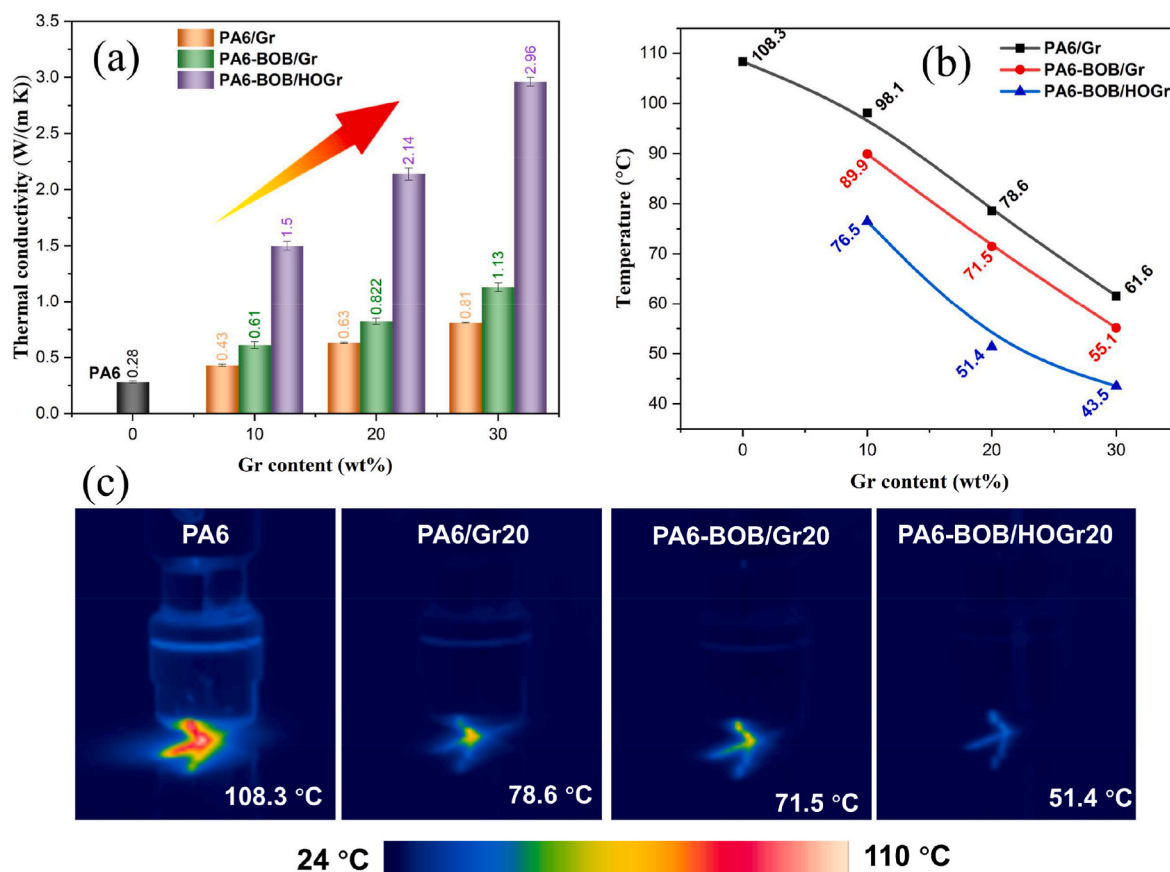


Fig. 8. (a) Thermal conductivity and (b) the maximum tribo-surface temperature during the friction of pure PA6 and its composites; (c) The infrared images of some representative samples during the friction test.

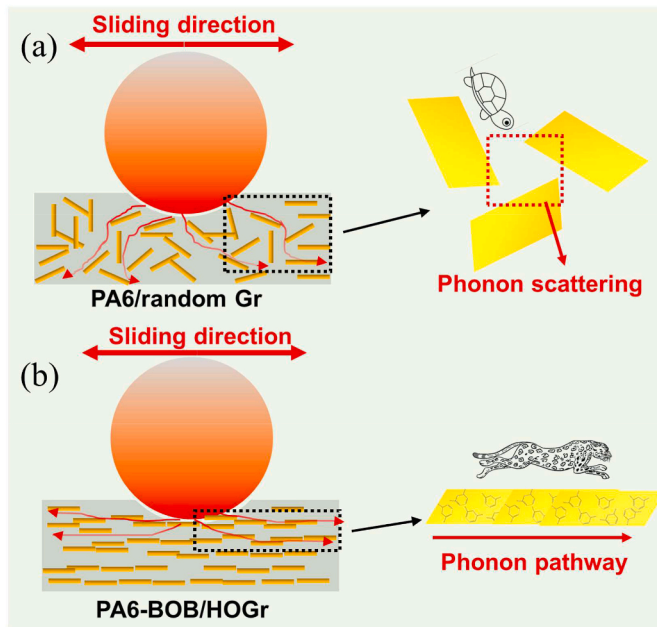


Fig. 9. Illustration of friction heat dissipation paths of composites during friction: (a) PA6/Gr and (b) PA6-BOB/HOGr.

and weak interfacial adhesion between Gr and PA6.

As shown in Fig. 10b, the better interfacial adhesion leads to a high tensile strength due to the more effective stress transfer at the interface of filler-polymer. The PA6-BOB/Gr20 exhibits the best strength value,

increasing about 51.9% compared with pure PA6. Furthermore, the highly oriented Gr in PA6 matrix results in higher strength, and the tensile strength of PA6-BOB/HOGr20 reaches 78 ± 4.7 MPa. This phenomenon is because the highly ordered Gr structure reduces the defects of the composite. Interestingly, elongations at the break of the multi-stage stretching extruded composites are always much more significant. The composites PA6-BOB/HOGr10 and PA6-BOB/HOGr20 show a ductile fracture, the elongation at break reached 217% and 150%, respectively. The brittle-to-ductile transition is also mainly attributed to homogeneous dispersion and high orientation of Gr in the PA6 matrix, which results in the defects being eliminated significantly. However, a ductile-to-brittle transition occurs when Gr loading increases to 30 wt% because high filler loading will lead to agglomeration.

SEM is used to further understand the effect of Gr orientation on mechanical properties to observe the fracture morphology of the composites after tensile failure. Many filaments (pointed by red arrows) are formed in the composite PA6-BOB/HOGr20 during stretching, shown in Fig. 11C1. The filaments can effectively dissipate energy, resulting in ductile fracture. However, the filaments did not appear in all the PA6/Gr, PA6-BOB/Gr composites, and the PA6-BOB/Gr composites with high loading of Gr. This phenomenon is caused by agglomerates of Gr or holes in the composites. The agglomerates will act as stress concentration sites and cause the formation of cracks during stretching, resulting in brittle fractures.

4. Conclusions

In this work, PA6/Gr composites with excellent wear resistance, ultrahigh heat dissipation capacity, and high ductility, were prepared successfully by incorporating BOB and highly oriented Gr flakes. The

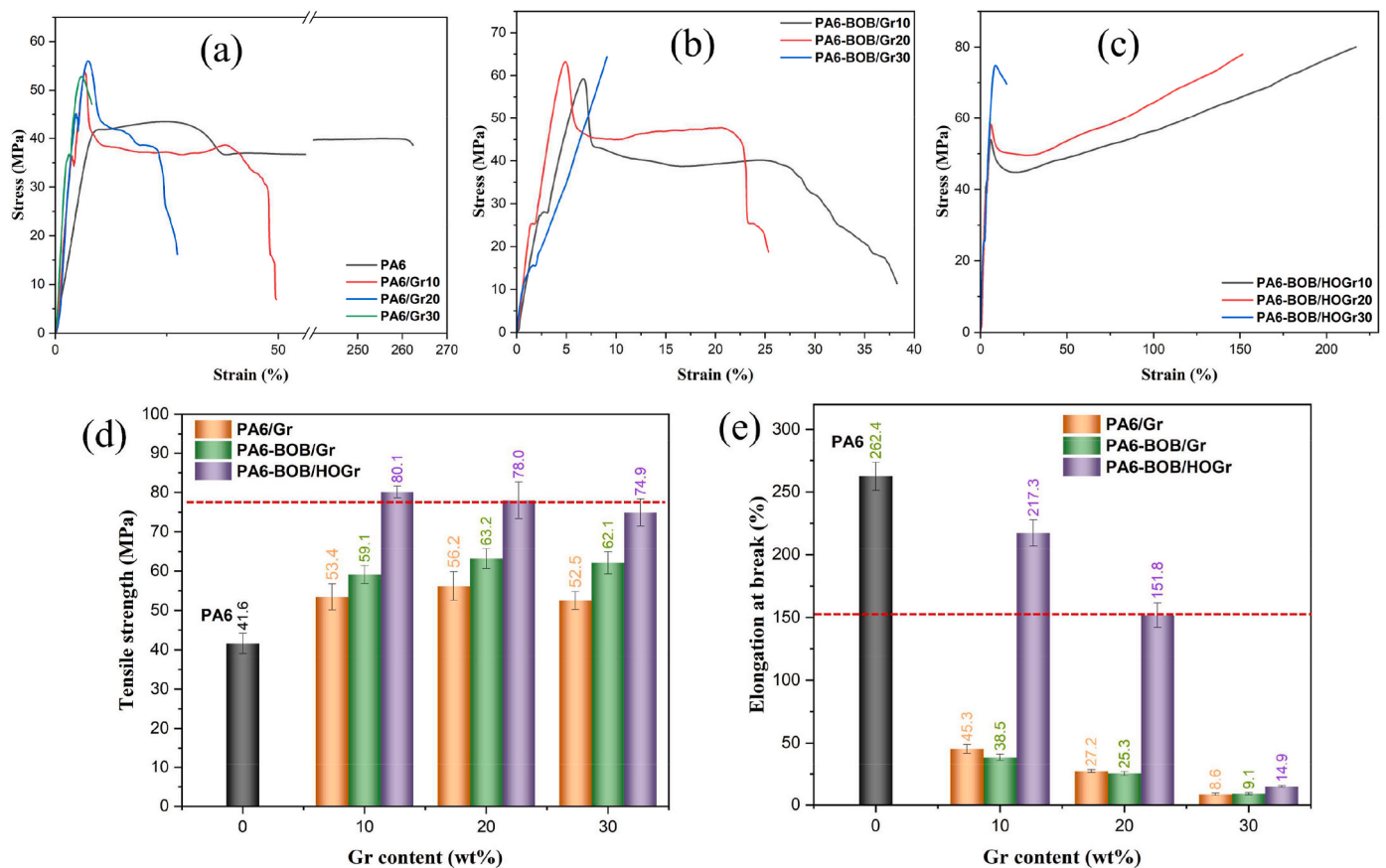


Fig. 10. Typical stress-strain curves of (a) pure PA6 and PA6/Gr10 ~ PA6/Gr30, (b) PA6-BOB/Gr10 ~ PA6-BOB/Gr30, (c) PA6-BOB/HOGr10 ~ PA6-BOB/HOGr30; statistical data of mechanical tests: (d) tensile strength and (e) elongation at break.

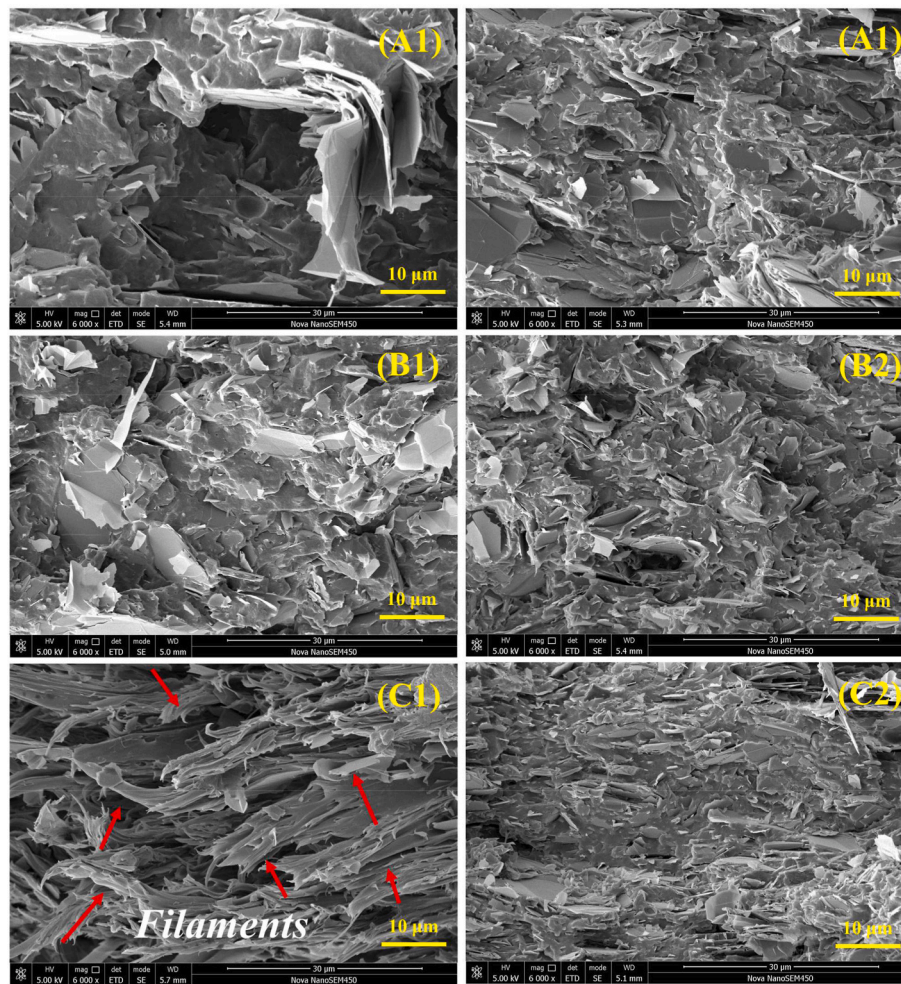


Fig. 11. SEM images of fracture morphology of some representative composites after stretching: (A1) PA6/Gr20, (A2) PA6/Gr30, (B1) PA6-BOB/Gr20 and (B2) PA6-BOB/Gr30, (C1) PA6-BOB/HOGr20, (C2) PA6-BOB/HOGr30.

COF and specific wear rate of the composites were considerably reduced with the addition of BOB, which can enhance the interfacial adhesion between Gr and PA6 through π - π interactions. In addition, the TC of PA6-based composites containing only 20 wt% highly oriented Gr reached as high as 2.14 W/(m K). As a result, the temperature of the tribo-surface can be controlled at a low level during friction. It is ascribed that the low COF reduces the generation of friction heat and the high TC dissipates the friction heat timely. Moreover, it still had extraordinary strength and ductility for PA6-BOB/HOGr20 with elongation at a break of 150% and tensile strength of 78.0 MPa. Therefore, this work provided new ideas for manufacturing advanced linear guide rail, liner, and linear bearing in industrial applications.

Author statement

Jing Chen: Investigation, Experiment, Data curation, Visualization, Writing - original draft, Writing - review & editing. Jiaming Zhu: Funding acquisition, Supervision. Yiyue Luo: Resources. Hong Wu: Conceptualization, Methodology, Formal analysis, Writing - review & editing, Supervision. Shaoyun Guo: Project administration, Supervision, Funding acquisition. Jianhui Qiu: Project administration, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compscitech.2022.109719>.

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